Object-Oriented System Analysis Design

2nd September 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

a) Define the following terms:

i) Cohesion **1**

Mark scheme
Cohesion is the focussing of a module on just one thing.

ii) An Actor in the context of use-case modelling. **1**

Mark scheme
An Actor is a person that performs an action depicted as a use-case.

iii) Cardinality or multiplicity **1**

Mark scheme
Cardinality or multiplicity is the number of items associated with an individual element.

b) *Multiple choice questions: Please select ONE (1) from the options provided*

i) Which of the following is NOT a UML diagram used in a system analysis model? **1**

A. use-case diagram
B. class diagram
C. dataflow diagram
D. state diagram

Mark scheme
C. data flow diagram

ii) Which of the following would NOT be depicted as an event in a UML state diagram? **1**

A. an input from a user
B. a time-out
C. a message from an object
D. a class

Mark scheme
D. a class

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Marks

iii) Which of the following items appears on a CRC card? **1**

A. class collaborators
B. class reliability
C. class responsibilities
D. Both A and B
E. Both A and C
F. Both B and C

Mark scheme
E. Both A and C

iv) Object Constraint Language (OCL) can be used for **1**

A. Specifying stereotypes and types in the class model
B. Describing pre- and post-conditions on Operations and Methods
C. Describing Guards
D. All the above
E. None of the above

Mark scheme
D. All the above

v) In object-oriented design, methods and attributes from a super class being available to a sub-class is called **1**

A. Cohesion
B. Inheritance
C. Decoupling
D. Prioritisation

Mark scheme
B. Inheritance

vi) In OO, a procedure or operation performed by a class is a **1**

A. Sub-class
B. Instantiation
C. Method
D. Attribute

Mark scheme
A. Method

vii) In Agile software processes, once requirements have been agreed, they **1**

A. cannot be changed
B. can be changed
C. can be changed once a running model of the system has been produced
D. can be changed once only

Mark scheme
B. Can be changed

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Marks

Total 10 Marks

Question 2

a) Discuss the considerations for choosing between Object-Oriented System Analysis and Design (OOSAAD) and Structured Systems Analysis and Design (SSADM). **2**

Mark scheme
SSADM and OOSAAD serve different purposes and have distinct methodologies. The choice between them depends on project requirements, stakeholder communication, and the desired quality of the final product.

Award 1 mark for each point, up to 2 marks.

b) Outline FOUR (4) differences between the OOSAAD and SSADM. **8**

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Marks

Mark scheme

	SSADM	***OOSAAD***
Focus	***emphasises processes involved in a software system. It models these processes as a series of connected steps. The emphasis is on process modelling.***	***OOSAAD centres around objects involved in a software system. It models these objects as instances of classes, encapsulating both data and behaviour.***
Process	***follows a top-down approach, breaking down complex systems into smaller, simpler parts for better understanding. Lends to life cycles that are step-wise, linear.***	***OOSAAD follows a bottom-up and/or top-down approach, building complex systems from smaller, understandable objects. Lends to life cycles that are iterative / incremental.***
Entity modelling	***focuses on the data flows between processes, emphasizing how data is manipulated.***	***OOSAAD focuses on modelling relationships and interactions among objects.***
Flexibility	***is valuable but can be rigid in following a structured approach. Its suitability depends on project requirements.***	***OOSAAD offers flexibility and adaptability, but its effectiveness depends on project context.***
Adoption	***mature and widespread – used for many legacy systems.***	***Contemporary (emerged in the late 90s).***

Award 2 marks for each comparison up to a maximum of 8 marks.

Total 10 Marks

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Marks

Question 3

a) Explain TWO (2) advantages and TWO (2) disadvantages of prototyping. **4**

Mark scheme

Advantages of prototyping in general

- ***Helps identify and test user requirements before the actual product creation.***
- ***Allows users to get a feel of the product, especially the higher the fidelity.***
- ***Enables active communication and follows through user needs***
- ***Tests and results in design, usability and maintainability improvements. Errors can be detected early.***
- ***Can also result in reduced development effort.***
- ***Iterative prototypes allow for flexibility of design.***
- ***Prototypes can be reused.***

Disadvantages of prototyping

- ***Can be time-consuming and expensive, especially when a throw-away prototype is created. Also, users may lose interest in the product making the activity a waste of time and money.***
- ***Can create challenges of implementing emerging ideas once prototype has been approved especially where a high-fidelity prototype has been created.***
- ***Iterative prototyping whilst good can result in inadequate documentation or create a huge documentation load.***
- ***After seeing a prototype, customers might expect earlier delivery than is feasible.***
- ***Prototyping may be limited by the prototyping tool and the prototype may not realise the full potential of the programming platform to be used for the actual product.***

2 marks for advantages total and 2 marks for disadvantages total.

b) Compare and contrast any TWO (2) of the prototyping methods below. **6**

i. Storyboards
ii. Wizard of OZ
iii. High level programming simulation.

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Marks

Mark scheme

Storyboards

Storyboards originated from the film industry, they are usually low fidelity (i.e. paper) prototypes but can also be created using a simulation with multimedia tools or a high-level programming mock-up. They are similar to use-case diagrams or state diagrams but with a focus on interface exhibits such as the layout, formats, and interaction design. The storyboard can be used as a basis for a Wizard of OZ run.

Wizard of OZ

Wizard of Oz technique: Could be considered to be a live storyboard and indeed the activity might be supported by a storyboard, use-case, or behavioural model. Only the user interface is developed using multimedia tools or high-level programming mock-ups like storyboards. Users interact with interface but messages are passed to a human who also responds when required. This is good for eliciting requirements.

High level programming simulation.

High level Programming Support/Simulation can be used to create storyboards and interfaces for wizard and Oz prototyping. The mock-ups are developed using Multimedia products (e.g. Shockwave; Hypercard; Toolbox) or simple high level languages such as 4GLs like Microsoft Access/Visual Basic or other code generator environments; a User Interface management system; screen painters; or reusable components.

For each of the two methods chosen, award 1 or 2 marks for description of the method; and 1 or 2 marks for comparison with the other method chosen – up to 3 marks each.

Total 10 Marks

Question 4

a) Differentiate between Blackbox and Whitebox testing. **2**

Mark Scheme

Whitebox testing is used for evaluating code functionality. Blackbox testing for evaluating the interface level. Blackbox testing is often referred to as surface testing, and purports to the testing structure, observable by end-users. Approaches also include deep structure testing which tests the internal structure of programs.

Award 1 mark for each point up to 2 marks

b) Describe TWO (2) aspects in which OO software testing and traditional non-object-oriented differ and give TWO (2) examples of OO Blackbox or Whitebox testing to help explain the difference. **8**

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Marks

Mark scheme

Any two differences from below:

Traditional Testing	*OO testing*
Unit testing is method by method i.e a unit is typically a subroutine, module or procedure.	*The smallest unit of testing is typically an encapsulated class or object. OO Whitebox testing focuses on methods of each class preferably focussing on one class at a time.*
Focuses on decomposition allowing testing of components in isolation.	*Focuses on composition allowing testing of a class/object, its attributes and operations.*
Emphasis is on testing system against predetermined criteria. Requires understanding of requirements and system specification.	*Emphasis is on isolated testing of classes/objects. Requires understanding of system's design and implementation.*
Testing examines data flow and algorithmic detail.	*Testing examines concrete and abstract state of the classes.*
Verifies behaviour of entire system: Integration testing checks that different components or modules can work together as expected.	*Verifies behaviour of each object/class: ensuring that objects/classes perform their intended functions and interactions to fulfil their relationships with other objects/classes.*
Integration testing is usually top-down as dictated by main controlling module.	*Integration testing can be top-down or bottom-up evaluation of groups of collaborating classes.*

Award up to 2 marks for each comparison (up to 4 marks).

Any two examples of OO Blackbox or Whitebox testing:

- ***Tactically, OO fault-based testing uses the inheritance structure to identify and test plausible defects that may be encountered in operation calls and messages.***
- ***In OO, partitioning can be based on object states, attributes or categories.***
    - ***Attribute-based partitioning classifies operations depending on how they use or modify an attribute.***

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- ***State-base partitioning differentiates between operations that cause state changes and those that do not.***
- ***Category-based partitioning groups methods according to the overarching function they perform: initialisation, computation, query, termination.***

- ***In OO, test cases simulate different scenarios or inputs that the object/ class might encounter in the real world. OO behavioural models (e.g sequence diagram based partitioning) and use-case diagrams provide added value for identifying the test cases and indeed ensure traceability and allowing testing of objects in every state.***
- ***Polymorphism also enhances the independence of each object, making them easier to test as standalone units. Orthogonal testing can be used to identify and minimise all possible scenarios.***
- ***In OO Integration Testing:***
  - ***thread-based testing tests all classes required to respond to one input or event***
  - ***use-based testing begins by testing independent classes and the dependent classes that make use of them***
  - ***cluster testing tests groups of collaborating classes for interaction errors***

Total 10 Marks

Question 5

a) Describe modular design and component-based design. **4**

Mark Scheme
Modular design and component-based design are closely related in that component-based design supports modularisation and vice versa. In modular design, the system is broken down into independent modules each of which or groups of which can form a component. (2 marks)

Component-based approach promotes reusability by allowing each component to be movable/transferable for reuse in a number of (sub-systems). It allows developers to concentrate on problem domain objects. Problem domain objects frequently recur so lend to reuse. Classes can be created on open-closed rule: both closed (stable and usable now) and open (extensible later). Reuse can involve whole objects or refactored objects. (2 marks)

b) Explain at least SIX (6) benefits of these approaches. **6**

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Marks

Mark scheme

Benefits of Modular design

- ***Functional independence: high cohesion and low coupling results in increased maintainability/change management, testability (2 marks for this if it includes cohesion and coupling below)***
  - ***Cohesion: promotes ability of a module to focus on just one thing as in the SOLID Single Responsibility Principle***
  - ***Coupling: degree to which a module is connected to other modules***
- ***Modular decomposability: systematic break down***
- ***Modular composability: supports reuse of existing modules***
- ***Modular understandability: module understood as a stand-alone unit***
- ***Modular continuity: minimisation of side-effects due to module changes***
- ***Modular protection: minimisation of side-effects due to processing errors***

Component-based approaches possess all advantages of modularisation plus:

- ***Time and Cost saving by reducing reinvention and resulting in faster delivery***
- ***Product reliability especially certified/inhouse components that have been pre-tested and sometimes already trialled***
- ***Proven solutions to common problems***
- ***Opportunity to focus on user requirements not coding***
- ***Increase consistency between applications***
- ***Afford modular design which lends to easier maintenance***

Award 1 mark for each benefit, up to 6 marks total for the question

Total 10 Marks

Question 6

a) Explain what is meant by a requirements catalogue in the context of system analysis and design. **2**

Mark Scheme
A requirements catalogue itemises, describes and prioritises the desired artefacts (functions and design features) of a system.

- ***1 mark for basic idea (eg list of system functions)***
- ***1 mark for further elaboration***

b) Describe the typical contents of a requirements catalogue. **8**

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Marks

Mark scheme

Award 1 mark for a basic point and additional mark for elaboration

- ***Requirement identifiers, description, rationale, processes and their inputs 1 mark***
- ***Viewpoints: requirements as perceived by different stakeholders. This would typically include requirements of all stakeholders and usage environments (generic requirements); those specific to individual stakeholders or environments; and how conflicting requirements have been resolved 2 marks***
- ***Purpose/rationale for each requirement, requirements sources – knowing the source can help to understand the requirements more (motives, intentions, purpose) 1 mark***
- ***Required system resources / special tools required to deploy requirement 1 mark***
- ***Operating system & other supporting hardware/software and explain how it might change over life of system***
  - ***Platform information e.g operating system versions, libraries***
  - ***Interface information e.g user agents, interface types***
  - ***Software dependencies***

  2 marks

Award 1 to 2 marks for each item listed up to 8 marks

Total 10 Marks

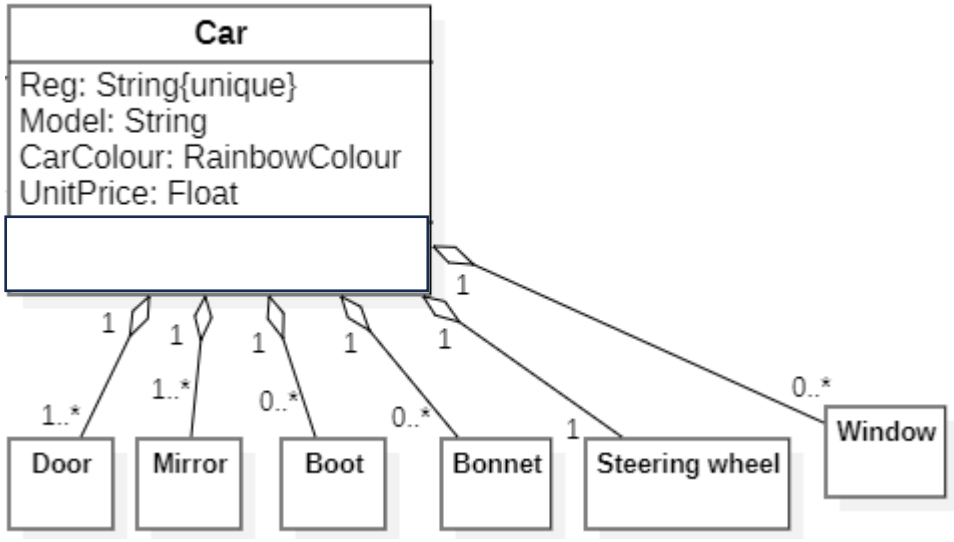
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Marks

Question 7

Draw a class diagram showing FOUR (4) only attributes of a car and SIX (6) only possible whole-part components. **10**

Mark scheme



Award up to 4 marks for car attributes

Award up to 3 marks for identification of components: 3 marks for 6 correct components, 2 marks for up to 5 correct components, 1 mark for up to 2 correct components.

Award 2 marks for showing relationships: 1 mark for up to 3 correct pairs, 2 marks for up to 6 correct pairs.

Award 1 mark for correct syntax

Total 10 Marks

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Marks

Question 8

Explain the FIVE (5) SOLID principles applicable to object-oriented (OO) design. **10**

Mark scheme

i. ***Single Responsibility Principle (SRP): Each class should have only one responsibility although sometimes this can result in a number of disjointed elements.***
ii. ***Open-closed-principle (OCP): Software units should be expandable without having to change their behaviour e.g inheritance and polymorphism help avoid data redundancy and unresponsiveness to change.***
iii. ***Liskov Substitution Principle (LSP): A derived class should always be used instead of its base class e.g age = dob – today()***
iv. ***Interface Segregation Principle (ISP): Many client specific interfaces are better than one - Interfaces should be perfectly adapted to the requirements of the accessing clients. This promotes multi-interface design which is key to inclusive design/support of diversity.***
v. ***Dependency Inversion Principle (DIP): Classes on a higher level of abstraction should never depend on classes on a lower level of abstraction. This guards against disjointed change impacts of subclasses that can result in repetition and loss of class cohesion.***

Students can also substitute any of the above by additional principles from Pressman:

- ***Release Reuse Equivalency Principle (REP): the granule of reuse is the granule of release.***
- ***Common Closure Principle (CCP): classes that change together belong together.***
- ***Common Reuse Principle (CRP): classes that cannot be used together should not be grouped together.***

Award 1 mark for identifying each principle and 1 mark for the description, up to 10 marks

Total 10 Marks

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Marks

Question 9

Compare and contrast the object-oriented analysis and design terms below. You must provide examples to illustrate your answers.

a) A Message and a Method **5**

Mark scheme

Award 1 mark for each valid point – the following are indicative points:

Action is initiated in OO programming by the transition of a message to an object responsible for the action. (1 mark) The message encodes the request for an action and is accompanied by any additional information (arguments) needed to carry out the request. (1 mark) The receiver is the object to whom the message is sent. If the receiver accepts the message, it accepts the responsibility to carry out the action. (1 mark)

Messages include keywords called a selector representing the method e.g. object myClock might have reportTime; resetTimeTo. Selectors may include one or more arguments or parameters e.g. resetTimeTo(newTime). (1 mark) In response to a message, the receiver will perform some action / method / operation / function / procedure to satisfy the request. (1 mark)

b) Abstract Class and Instance of a class **5**

Mark scheme

Award 1 mark for each valid point – the following are indicative points:

An Abstract Class is a class for which there are no direct instances (1 mark) and is usually only created to group subclasses e.g. Mammal, it is used only to create subclasses. (1 mark)

An Instance of a class is a child object created (instantiated) from a class. (1 mark) The instance is expected to behave in certain ways and have attributes that are consistent with the class e.g. a dog is an instance of the mammal class which is an instance of the animal class. (1 mark) Another instance of the class, say a bird, will manifest the same attributes and behaviours that define it as an animal. (1 mark)

Total 10 Marks

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Marks

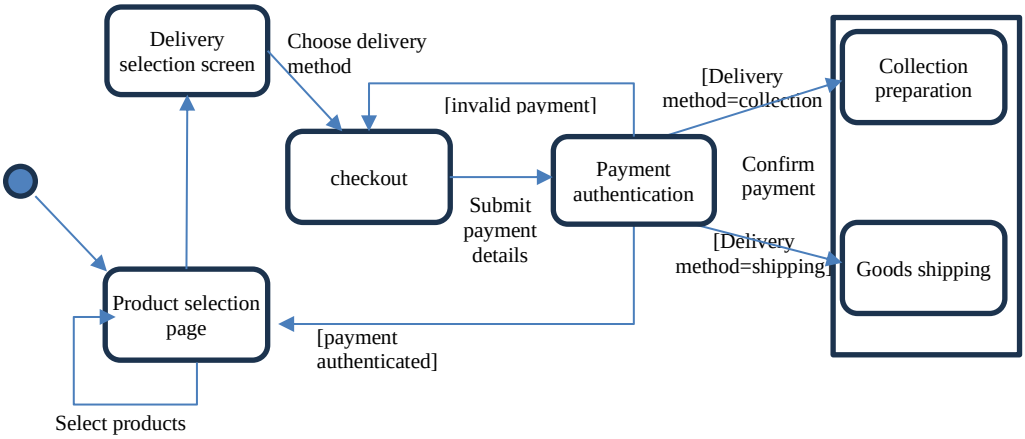
Question 10

Draw a UML State Diagram to illustrate the following process for an online purchase and delivery of products. **10**

1. Customers select one or more products from a product selection page.
2. Customers select checkout and enter payment details.
   a. If payment is not valid, customers are returned to the Checkout page
   b. If payment is valid, customers select a delivery method (collection or shipping) and complete appropriate details.
3. The system confirms the order and warehouse fulfils the order.

Your response must also demonstrate use of guards, nesting and self-messaging.

Mark scheme



The diagram is one example, other valid representations should be awarded full marks. An equally valid / complete response could have less or more states/ transitions, the key thing is validity.

Award up to 3 marks for states and up to 3 marks for transitions for correct representation. Award 2 marks for each where it is incomplete but mostly correct and 1 mark for each where there are some valid states/transitions.

Award 1 mark for use of nesting
Award 1 mark for correct use of a guard, 2 marks for consistent & correct use of guards
Award 1 mark for self-messaging

Total 10 Marks

End of paper

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Marks

Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1 - 4	Yes
2	2	Yes
3	1, 4	Yes
4	2, 4	Yes
5	1, 4	Yes
6	1	Yes
7	2, 3	Yes
8	2, 4	Yes
9	2	Yes
10	3	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
1. Discuss various approaches to systems analysis and design and explain their strengths and weaknesses.	Unable to identify and assess approaches.	In an extremely limited way, can identify and assess methodologies and methods to reach limited solutions.	Can adequately identify and assess appropriate methodologies and methods to reach appropriate solutions.	Can sufficiently identify and assess appropriate methodologies and methods to reach good solutions.	Can coherently identify and assess coherent methodologies and methods to reach well-defined and structured solutions.
2. Apply object-oriented techniques for systems analysis and modelling of well-defined but complex problems.	Unable to explain key principles or discuss developments.	In an extremely limited way, can explain principles and discuss developments.	Can adequately explain principles and discuss developments.	Has very good grasp of explain principles and discuss developments.	Can coherently discuss developments and explain and adapt principles.
3. Explain and use UML to represent object-oriented models.	Little to no understanding of UML or object-oriented models.	Limited understanding of UML or object-oriented models.	Adequate ability to apply UML with sufficient representations.	Demonstrates very good ability to apply and adapt UML models with a substantial system understanding.	Outstanding ability to apply UML and adapt models in a manner that demonstrates understanding of effectiveness of methods, actions and results.
4. Understand the integration between the design models and implementation.	Lacks awareness of different perspectives or approaches within the area of study.	Has extremely limited awareness of different perspectives or approaches within the area of study.	Has adequate awareness of different perspectives or approaches within the area of study. Understands requirements traceability but with some gaps.	Has very good, informed awareness of different perspectives or approaches within the area of study. Understands requirements traceability.	Excellent awareness of different perspectives or approaches within the area of study.

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